Endogenous Firm Scope via Organizational Complexity

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Scalable Capability, Bounded Scope

- The rise of intangibles makes capability easy to scale: build the blueprint once, deploy it everywhere at near-zero cost. (Argente, Moreira, Oberfield & Venkateswaran 2025; Crouzet, Eberly, Eisfeldt & Papanikolaou 2024)
- Yet realized scope is bounded and related: firms add nearby markets, mostly through R&D and acquisitions. (Hoberg & Phillips 2025)
- Even Amazon (healthcare) and Google (hardware) stall in distant markets. If capability replicates so freely, why doesn't a handful of firms span the whole economy?

What bounds firm scope?

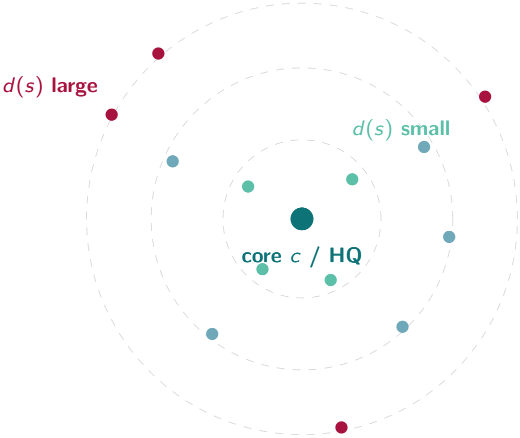
This Paper: Scope Bounded by Complexity, Not Count

The binding constraint is not the number of markets a firm serves but how different they are. We capture this as organizational complexity: the cumulative heterogeneity across a firm's footprint.

- Distant markets are organizationally heterogeneous; adapting the non-rival blueprint there consumes scarce headquarters attention, so capability does not translate into scope.
- Expansion is non-separable: adding a market raises the cost of serving every other.
  (unlike count-based theories: span of control, Lucas 1978; entry costs, Melitz 2003; knowledge hierarchies, Garicano 2000, Garicano & Rossi-Hansberg 2006)
- Against the winner-take-all intuition, the model predicts local dominance need not aggregate: scalable capability need not concentrate the economy.

Environment

A Firm in One Picture



- One non-rival blueprint, created at the core, deployed across a footprint of heterogeneous markets.
- Each market sits at organizational distance $d(s)$ from the core.
- Deploying to distant markets must be adapted; adaptation consumes rival headquarters attention.

Markets and Heterogeneity

market space & mass	(S, Σ, μ) with core $c \in S$
what each market is like	$\xi(s) \in \mathbb{R}^k$
organizational distance	$d(s) = \delta\big(\xi(s), \xi(c)\big) \in \mathbb{R}_+$

relative to core, via δ

μ records the mass of markets, ξ records what they are, and $d(s)$ is a market's organizational distance from the core. Each market also carries exogenous capacity $\bar{X}(s) > 0$.

Why layered: μ separates how many markets from how different they are, which a single normed space would fuse; and δ encodes organizational technology, not required to be a norm.

▶ details

Organizational Complexity

Integrating organizational distance over the footprint gives a single firm-level state:

Definition. For any measurable footprint $M \subseteq S$,

$$C(M) \equiv \int_M d(s)\,\mu(ds).$$

Complexity is driven by cumulative heterogeneity, not the number of markets.

▶ details

The Blueprint

A firm operates a centralized blueprint $z \geq 0$, created at headquarters:

- Non-rival: one blueprint z serves all markets at once.
- Centralized: created and maintained at headquarters.
- Adapted: each deployment must be customized locally.

Creating quality z requires headquarters attention

$$z = \Gamma(h_z) \qquad \Longleftrightarrow \qquad \chi(z) \equiv \Gamma^{-1}(z) \text{ units of attention,}$$

with Γ continuous, strictly increasing, $\Gamma(0) = 0$.

Deployment and Returns

Effective deployment of the blueprint in market s:

$$z_s \;=\; \theta\big(\underset{+}{a(s)},\; \underset{-}{d(s)}\big)\, z, \qquad a(s) \;=\; A\big(\underset{+}{h(s)},\; \underset{+}{\ell(s)}\big)$$

θ: portability A: adaptation

Effective deployment and local input generate the market-level payoff:

$$R\big(\underset{+}{z_s},\; \underset{+}{x(s)}\big)$$

$h(s)$: headquarters attention $\ell(s)$: local effort $x(s)$: local production input

▶ details

The Central Friction: Rival Headquarters Attention

Headquarters attention is rival within the firm: split between blueprint creation and adaptation. Serving a more heterogeneous footprint makes it more expensive everywhere.

Cost of headquarters attention:

$$q(C(M))\left(\underbrace{\chi(z)}_{\text{creation}} + \underbrace{\int_M h(s)\,\mu(ds)}_{\text{adaptation}}\right)$$

Organizational congestion function:

- $q : \mathbb{R}_+ \to \mathbb{R}_+, \;\; q(0) = 1, \;\; q' > 0, \;\; q(C) \to \infty.$
- q is an ordinary congestion wedge. The economic content is the state it congests on: cumulative heterogeneity $C(M)$ (not size!).

▶ details

The Firm's Problem

The firm chooses blueprint quality z, footprint M, and allocations $\{x(s), \ell(s), h(s)\}$:

$$\max_{z,\, M,\, \{x,\ell,h\}} \; \int_M R\big(z_s, x(s)\big)\, \mu(ds) \;-\; q\big(C(M)\big)\Big(\chi(z) + \int_M h(s)\, \mu(ds)\Big)$$

$$\text{s.t.} \;\; x(s) + \ell(s) \leq \bar{X}(s) \quad \forall\, s \in M$$

$$\text{where} \;\; z_s = \theta\big(a(s), d(s)\big)\, z$$

$$M = \{\, s : x(s) + \ell(s) + h(s) > 0 \,\}$$

Baseline Results

Endogenous Scope Limit

Fix capability z. What bounds the firm's reach?

Proposition (bounded complexity). At any (minimal-support) optimum,

$$C\big(M^\star(z)\big) = \int_{M^\star(z)} d(s)\,\mu(ds) \; < \; \infty.$$

- Bounded cumulative heterogeneity but not the mass $\mu(M^\star)$, which may be infinite.
- A finite mass of materially distinct markets ($d \geq \tau$); an unbounded mass of near-core ones.

Expansion Is Non-Separable

Let $\Pi_s(z; C)$ be the value of operating market s at complexity C.

Proposition (organizational congestion). $\Pi_s(z; C)$ is weakly decreasing in C; hence adding markets Δ weakly lowers every incumbent:

$$\Pi_s\big(z; C(M \cup \Delta)\big) \;\leq\; \Pi_s\big(z; C(M)\big) \quad \forall\, s \in M.$$

- A market's value depends on the whole footprint, through C. Thus selection is non-separable.
- Separable theories (Lucas–Garicano–Melitz) cannot generate this.

Related Expansion: A Cutoff in Heterogeneity

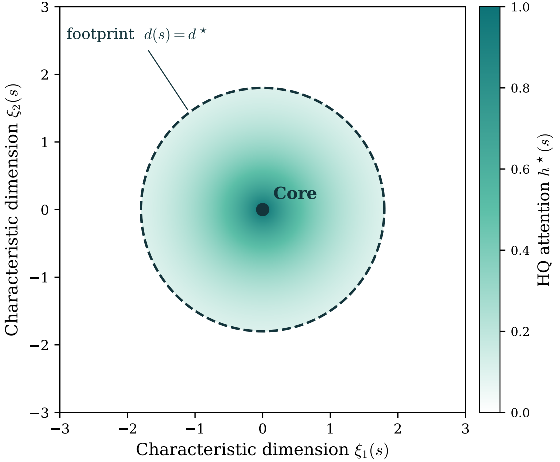
Which markets does the firm serve?

Proposition (cutoff selection). Under a heterogeneity-ordering condition, there is a threshold $d^\star$ such that the firm serves exactly the markets within it:

$$M^\star(z) = \{\, s : d(s) \leq d^\star \,\}.$$

- Firms serve markets near the core and exclude those beyond the threshold $d^\star$ (related expansion, Hoberg & Phillips 2025).
- No fixed costs or decreasing returns: selection comes from organizational scarcity.
- Threshold rests on a heterogeneity-ordering condition, e.g. uniform local capacity.
  - ▶ details

Core–Periphery Organization



$h^\star(s)$ concentrates at the core, a gradient from organizational scarcity.

▶ details

Endogenous Blueprint: Returns to Scale

Now let the firm choose blueprint quality z. The blueprint is non-rival. How does scope shape the incentive to invest in it?

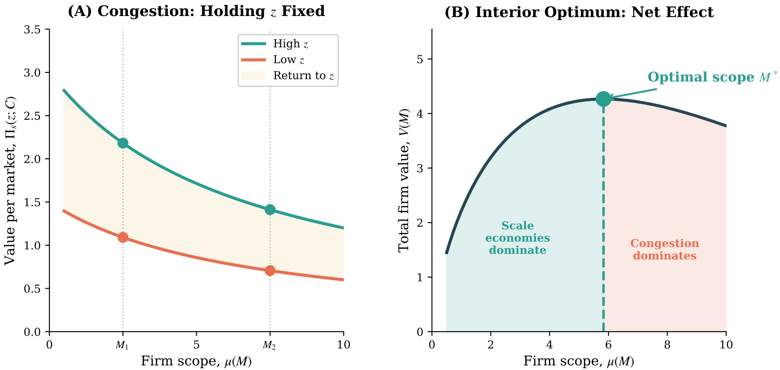
Proposition (returns to blueprint scale). For any $z_2 \geq z_1$ and $M_1 \subseteq M_2$,

$$\int_{M_2} \left[\Pi_s(z_2; C) - \Pi_s(z_1; C)\right] \mu(ds) \ \geq \ \int_{M_1} \left[\Pi_s(z_2; C) - \Pi_s(z_1; C)\right] \mu(ds).$$

- The return to a better blueprint rises with the served set. Every market deploys the same blueprint, amplifying the gain from improving it.
- Scope and blueprint quality are complements: a wider footprint $\Rightarrow$ stronger incentive to raise z.

Two Forces Shape Firm Scope

Expanding the footprint $M \to M' = M \cup \Delta$ pits two forces against each other:



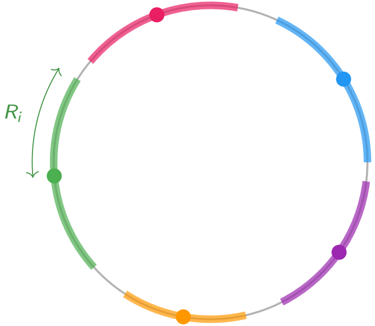
$$\Delta_\Pi = \underbrace{\left[\Pi_s(z^*(M'); C(M')) - \Pi_s(z^*(M); C(M'))\right]}_{\text{blueprint adjustment } (\geq 0)} + \underbrace{\left[\Pi_s(z^*(M); C(M')) - \Pi_s(z^*(M); C(M))\right]}_{\text{organizational congestion } (\leq 0)}$$

Local Dominance & Concentration

What the single-firm scope limit implies across many firms.

A Multi-Firm Environment

Embed many firms on a shared space: the torus $\mathbb{T}_L$, a circle of circumference L.



- Core c_i (**dot**) and footprint $M_i = \{s : \delta(s, c_i) \leq R_i\}$ (**arc**).
- Scope radius $R_i = \mathcal{R}(z_i)$ (optimal reach), bounded by organizational complexity.
- Each firm dominates locally near its core.

Does this local dominance add up to aggregate concentration?

▶ details

Local and Aggregate Concentration: One Object

Firm i's local activity share is $\omega_i(s)$. The overlap kernel:

$$\mathcal{O}(x) = \frac{1}{L}\int_{\mathbb{T}_L} \sum_i \omega_i(s)\,\omega_i(s+x)\,ds$$

Proposition (overlap-kernel identity).

$$\mathcal{O}(0) \;=\; \overline{\mathrm{HHI}}_{\mathrm{loc}} \;\equiv\; \tfrac{1}{L}\int_{\mathbb{T}_L} \textstyle\sum_i \omega_i(s)^2\,ds,$$

$$\tfrac{1}{L}\int_{\mathbb{T}_L} \mathcal{O}(x)\,dx \;=\; \mathrm{HHI}_{\mathrm{agg}} \;\equiv\; \textstyle\sum_i \bar{\omega}_i^2.$$

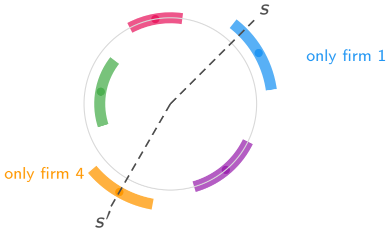
where $\bar{\omega}_i = \frac{1}{L}\int_{\mathbb{T}_L}\omega_i$ is the average share, and HHI is the Herfindahl–Hirschman index.

- How fast $\mathcal{O}$ decays governs the gap between local and aggregate concentration.

▶ details

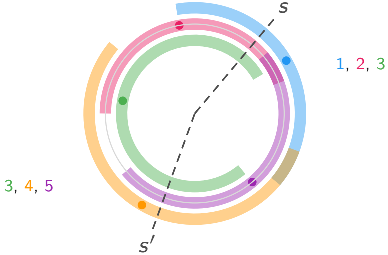
Scope Compression Lowers Overlap

Organizational complexity



compressed arcs, little overlap

Without scope compression



wide arcs, same firm spans s, s'

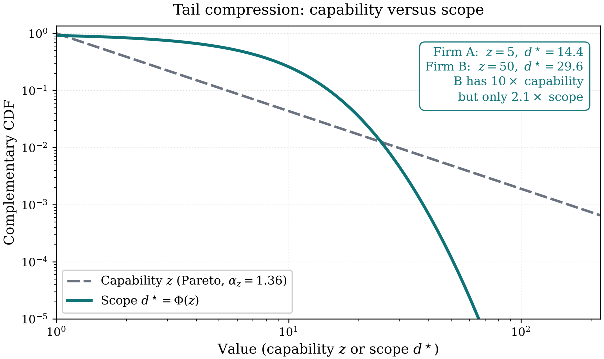
- Compressed scope $\Rightarrow \mathcal{O}$ decays fast: each firm dominates near its core, with little overlap elsewhere.
- Local dominance does not aggregate: high local HHI, low aggregate HHI. Scalable capability need not concentrate the economy.

▶ details

Quantitative Illustration

Tail Compression and Missing Giants

Calibrate the model to Hoberg–Phillips scope: (mean 7.6, median 6.0, sd 5.8) with blueprints z distributed according to Pareto. ▶ details

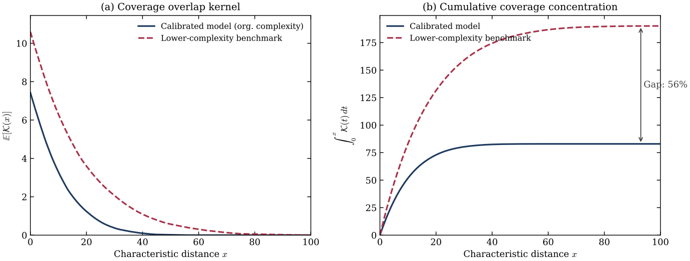


Pareto capability ($\alpha_z = 1.36$) $\Rightarrow$ a thinner-tailed scope distribution: $10\times$ capability buys only $\sim 2.1\times$ scope: the "missing giants."

Local Dominance, Bounded Aggregate Concentration

Embed the calibrated radii on a torus and compare with a lower-complexity benchmark.

▶ details



$\mathcal{K}(x)$ is the coverage kernel: same-firm presence at distance x, the extensive-margin counterpart of $\mathcal{O}$.

Complexity makes coverage decay faster: integrated overlap is **56% lower**.

Local dominance does not aggregate: no firm spans enough distant markets.

Key Takeaways

1. Scope is bounded by organizational complexity: the cumulative heterogeneity of a firm's markets, not their count.
2. Expansion is non-separable: scope is the net of blueprint scalability and organizational congestion.
3. Capability does not buy scope one-for-one: adapting to heterogeneous markets compresses the scope tail, the missing giants.
4. Local dominance need not aggregate: a firm dominates near its core while aggregate concentration stays bounded.

Appendix: Technical Assumptions

Assumption: Non-atomic Market Space

◀ back

Assumption (Non-atomic market space). (S, Σ, μ) is non-atomic: for every $E \in \Sigma$ with $\mu(E) > 0$ and every $\varepsilon \in (0, \mu(E))$, there exists $E' \subseteq E$, $E' \in \Sigma$, with $\mu(E') = \varepsilon$.

Rules out indivisible markets of positive mass: the firm can expand or contract its footprint by arbitrarily small amounts. The market space behaves like a continuum (Lebesgue on an interval), the natural idealization for scope spanning many small, related activities.

Assumption: Positive Heterogeneity

◀ back

Assumption (Positive heterogeneity). $d(s) > 0$ for μ-a.e. $s \in S$.

Rules out a positive-mass cluster of markets identical to the core. The normalization $d(c) = 0$ is consistent with this since the core is a null reference point. The firm may still operate across vast collections of nearly core-aligned markets with arbitrarily small $d(s)$.

Example (organizational distance). A natural form is a weighted norm,

$$d(s) = \left(\sum_{j=1}^{k} \omega_j \, |\xi_j(s) - \xi_j(c)|^p\right)^{1/p},$$

but no result requires $d(\cdot)$ to satisfy the axioms of a norm.

Assumptions: Deployment and Effectiveness

◀ back

Assumption (Adaptation and portability). $A : \mathbb{R}_+^2 \to \mathbb{R}_+$ continuous, weakly increasing in each argument. $\theta : \mathbb{R}_+ \times \mathbb{R}_+ \to [0,1]$ continuous, weakly increasing in a, weakly decreasing in d, with $\lim_{a\to\infty} \theta(a,d) = 1$ for all $d \geq 0$.

Assumption (No free deployment). For any s with $d(s) > 0$,

$$\theta\big(A(0,\ell),\, d(s)\big) = 0 \quad \text{for all } \ell \geq 0.$$

Assumption (Blueprint effectiveness). $R(z_s, x)$ weakly increasing in each argument.

An Inada-type portability technology: zero deployment without headquarters attention, full deployment in the limit of unbounded adaptation.

Assumptions: Congestion and Integrability

◀ back

Assumption (Organizational congestion). $q : \mathbb{R}_+ \to \mathbb{R}_+$ with $q(0) = 1$; continuously differentiable and strictly increasing; $\lim_{C\to\infty} q(C) = \infty$.

Broader, more heterogeneous scope raises coordination burdens, increasing the marginal cost of headquarters attention anywhere in the firm, including blueprint creation $\chi(z)$.

Assumption (Conditional integrability). For fixed blueprint quality z, gross payoffs are dominated by some $\bar{R}_z \in L^1(S, \mu)$, ruling out unbounded payoffs at fixed technology.

Appendix: The Heterogeneity-Ordering Condition

◀ back

Assumption (Heterogeneity ordering). For any fixed (z, C), the market value $\Pi_s(z; C)$ is weakly decreasing in $d(s)$:

$$d(s_1) \leq d(s_2) \implies \Pi_{s_1}(z; C) \ \geq \ \Pi_{s_2}(z; C).$$

More heterogeneous markets are weakly less valuable to serve; this monotonicity is what turns selection into a threshold in d.

Sufficient primitive condition. Uniform local capacity $\bar{X}(s) = \bar{X}$: the inner feasible set is common across markets, effective deployment $\theta(A(h, \ell), d(s))\, z$ falls in d, so the objective $R(z_s, x) - q(C)\, h$ is pointwise larger at smaller d; taking suprema delivers the ordering.

Sufficient, not necessary: the ordering also holds for non-uniform $\bar{X}$ co-varying suitably with d. It restricts neither returns to scale nor within-market technology.

Proposition: Core–Periphery Allocation

◀ back

Single-crossing (decreasing differences). With $V(h, d; z) \equiv \sup_{x+\ell\leq\bar{X}} R\big(\theta(A(h,\ell), d)\, z,\ x\big)$, for $h_1 \geq h_0$ and $d_1 \leq d_2$,

$$V(h_1, d_1; z) - V(h_0, d_1; z) \ \geq\ V(h_1, d_2; z) - V(h_0, d_2; z).$$

Proposition (core–periphery allocation). Under single-crossing, constant local capacity $\bar{X}(s) = \bar{X}$, and single-valued $h^{\star}(s)$: for any $s_1, s_2 \in M$,

$$d(s_1) \leq d(s_2) \implies h^{\star}(s_1) \ \geq\ h^{\star}(s_2).$$

Headquarters attention is weakly decreasing in heterogeneity: the incremental payoff from attention is higher in low-d (core) markets.

Appendix: The Torus Environment

◀ back

Characteristic space. Torus $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$ of circumference L, with torus metric $\delta(s, s')$.

Firms. Firm i: core c_i, blueprint z_i. By the cutoff result (Prop 2.3) the optimal footprint is the arc

$$M_i = \{s \in \mathbb{T}_L : \delta(s, c_i) \leq R_i\}, \qquad R_i = \mathcal{R}(z_i) = \Phi(z_i)/2,$$

with scope radius induced by the scope function Φ of the distribution section.

No-double-overlap: $R_i \leq L/4$: arcs short enough that two shifted copies overlap only on the nearest side. **Cores** $\{c_i\}$ are exogenous (no entry, exit, or placement modeled).

Appendix: Local Shares and the Two Kernels

◄ back

Local share (zero-activity convention): $\omega_i(s) = y_i(s)/Y(s)$ if $Y(s) > 0$, else 0; $\sum_i \omega_i(s) \in \{0, 1\}$.

$$\mathrm{HHI}(s) = \textstyle\sum_i \omega_i(s)^2, \quad \overline{\mathrm{HHI}}_{\mathrm{loc}} = \frac{1}{L}\int \mathrm{HHI}, \quad \mathrm{HHI}_{\mathrm{agg}} = \sum_i \bar{\omega}_i^2, \;\; \bar{\omega}_i = \frac{1}{L}\int \omega_i.$$

Two kernels. Share-weighted $\mathcal{O}$ (genuine HHI, Prop 4.1) vs. coverage $\mathcal{K}$ (extensive-margin co-presence):

Coverage formula (Prop 4.2). For interval footprints with $R_i \leq L/4$,

$$\mathcal{K}(x) = \frac{1}{L}\textstyle\sum_i (2R_i - x)_+, \qquad \mathbb{E}[\mathcal{K}(x)] = \frac{N}{L}\,\mathbb{E}\big[(2\mathcal{R}(Z) - x)_+\big].$$

The calibration's $\mathbb{E}[\mathcal{K}(x)]$ (and the 56% gap) is the extensive-margin object; the genuine-HHI statement runs through the share-weighted $\mathcal{O}$.

Appendix: Scope Compression Reduces Overlap

◄ back

Proposition (scope compression reduces coverage overlap). If $\mathcal{R}(z) \le \widetilde{\mathcal{R}}(z) \le L/4$ (F_z-a.e., strict on a positive-measure set), then

$$\mathbb{E}[\widetilde{\mathcal{K}}(x)] \;\ge\; \mathbb{E}[\mathcal{K}(x)] \;\; \forall x, \qquad \mathbb{E}\Big[\tfrac{1}{L}\int_0^{L/2} \widetilde{\mathcal{K}}\Big] \;>\; \mathbb{E}\Big[\tfrac{1}{L}\int_0^{L/2} \mathcal{K}\Big].$$

Appendix: Calibration

◀ back

	Target (HP)	Model
Targeted moments		
Mean scope	7.55	7.36
Median scope	6.00	6.09
Std. dev. of scope	5.80	5.77
Calibrated parameters		
Pareto tail index α_z	—	1.36
Portability decay λ	—	18.11
Congestion weight κ	—	0.005
Congestion elasticity φ	—	1.96
Non-targeted		
P90 / P99 scope	—	15.0 / 26.1
Skewness	—	1.47

$H(d) = \exp(d/\lambda)(1 + \kappa d^2/2)^{\varphi}$, Pareto z. **Caveat:** 3 targets, 4 parameters: a disciplined illustration of magnitudes, not separately identified primitives.

Appendix: The Lower-Complexity Benchmark

◀ back

Construction. Both economies share the calibrated capability distribution and the same torus ($L = 300$, uniform cores). They differ only in the congestion elasticity φ of the scope technology:

$$H(d) = \exp(d/\lambda)\,(1 + \kappa d^2/2)^{\varphi}, \qquad R = \mathcal{R}(z) = \Phi(z)/2, \quad \Phi = H^{-1}.$$

- Calibrated economy: $\varphi = 1.96$, so organizational complexity bites.
- Lower-complexity benchmark: $\varphi_{\mathrm{low}} = 0.5$, a weaker congestion penalty, so scope radii are larger and less compressed.

A lower-complexity benchmark, not frictionless linear scaling: scope is still capability-bounded, only the complexity penalty is softened. The 56% is the gap in integrated coverage $\frac{1}{L}\int_0^{L/2}\mathcal{K}$ between the two economies.